

JEE MAIN REVISION MASTER: INDEFINITE INTEGRALS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 18 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Standard Integrals, Substitution & By Parts	9 PYQs	Integration Techniques
High	Partial Fractions & Special Algebraic Integrals	9 PYQs	Algebraic Integrals

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Standard Integrals, Substitution & By Parts	$\int e^x [f(x) + f'(x)] dx = e^x f(x) + C$, $\int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \ln \left \frac{x-a}{x+a} \right + C$, $\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \left(\frac{x}{a} \right) + C$
Partial Fractions & Special Algebraic Integrals	$\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + C$, $\int \frac{dx}{\sqrt{x^2 \pm a^2}} = \ln x + \sqrt{x^2 \pm a^2} + C$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Standard Integrals, Substitution & By Parts (High Yield — 9 PYQs)

Core Principle: Fundamental integrals, substitution methods, ILATE integration by parts, and linear/quadratic partial fractions.

■ **Caution / JEE Pitfall:** In integration by parts, follow the ILATE priority strictly for choosing the first function u (Inverse, Logarithmic, Algebraic, Trigonometric, Exponential).

■ **Speed Trick:** The pattern $\int e^x [f(x) + f'(x)] dx = e^x f(x) + C$ appears with high frequency in both JEE Main and KCET.

Representative Source PYQs:

- [2025 Evening Shift] Q#2 (Numerical): "If $\int \left(\frac{1}{x} + \frac{1}{x^3} \right) (\sqrt{23}) dx = -\frac{\alpha}{3(\alpha+1)} (3x^\beta \ln \beta) + x^\gamma \left(\frac{\alpha+1}{\alpha} \right) + C, x > 0$, (a...)" [Ans: 19]
- [2025 Evening Shift] Q#4 (Numerical): "If $\int \frac{2x^2 + 5x + 9}{(x^2 + x + 1) \sqrt{x^2 + x + 1}} dx = \alpha \sqrt{x^2 + x + 1} + \beta \log_e(x + 1/2) + \sqrt{x^2 + x + 1} + C, \dots$ " [Ans: 16]
- [2024 Evening Shift] Q#2 (Numerical): "If $\int \operatorname{cosec}^5 x dx = \alpha \cot x \operatorname{cosec} x (\operatorname{cosec}^2 x + (3/2)) + \beta \log_e(x) |\tan(x/2)| + C$ where $\alpha, \beta \in \mathbb{R}$ and C is the c..." [Ans: 1]

3.2 Partial Fractions & Special Algebraic Integrals (High Yield — 9 PYQs)

Core Principle: Integration using partial fraction decompositions, linear over quadratic forms, and standard quadratic root substitutions.

■ **Caution / JEE Pitfall:** Check if degree of numerator $\geq$ degree of denominator; if so, polynomial long division **MUST** be performed first.

■ **Speed Trick:** For integrals of type $\int (dx / a + b \cos^2 x)$, divide numerator and denominator by $\cos^2 x$ and substitute $\tan x = t$.

Representative Source PYQs:

- **[2026 Morning Shift] Q#1 (Numerical):** "If $\int \frac{(\sin x)^{-\frac{11}{2}} (\cos x)^{-\frac{5}{2}}}{dx} = -\frac{1}{\frac{p_1}{q_1} (\cot x)^{\frac{c_9}{2}} - \frac{1}{\frac{p_2}{q_2} (\cot x)^{\frac{5}{2}} - \frac{1}{\dots}}$ " **[Ans: 16]**
- **[2025 Evening Shift] Q#3 (Numerical):** "If $\int \frac{(\sqrt{1+x^2}+x)^{10} (\sqrt{1+x^2}-x)^9 dx}{m} ((\sqrt{1+x^2}+x)^n (n\sqrt{1+x^2}-x)) + C$ where C is the c..." **[Ans: 379]**
- **[2024 Evening Shift] Q#1 (Numerical):** "If $\int \frac{1}{\sqrt{5}} dx = A((\alpha x - 1 / \beta x + 3))^B + C$, where C is the constant of integration, then the value of α ..." **[Ans: 7]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 22nd January Morning Shift (Q#1)

If $\int (\sin x)^{-11/2} (\cos x)^{-5/2} dx = -\frac{1}{p_1} q_1 (\cot x)^{9/2} - \frac{1}{p_2} q_2 (\cot x)^{5/2} - \frac{1}{p_3} q_3 (\cot x)^{1/2} + \frac{1}{p_4} q_4 (\cot x)^{-3/2} + C$, where p_i and q_i are positive integers with $\gcd(p_i, q_i) = 1$ for $i = 1, 2, 3, 4$ and C is the constant of integration, then $\frac{1}{15p_1 p_2 p_3 p_4 q_1 q_2 q_3 q_4}$ is equal to

Correct Answer: 16

Solution Analysis: $\int (\sin x)^{-11/2} (\cos x)^{-5/2} dx = \int \frac{1}{\tan^{11} x} \cdot \sec^8 x dx = \int \frac{(1 + \tan^2 x)^3 \sec^2 x \tan^{11} x dx}{\tan^{11} x}$
 put $\tan x = t$ $\frac{1}{5} \left(\frac{1}{5} \right)^2 \left(\frac{2}{3} \right)^2 \left(\frac{4}{1} \right)^2 \left(\frac{2}{3} \right)^2 \left(\frac{4}{1} \right)^2 = \frac{15 \times 2 \times 6 \times 6 \times 2}{9 \times 5 \times 1 \times 3} = 16$

Benchmark Problem 2: JEE Main 2025 (Online) 7th April Evening Shift (Q#2)

If $\int \left(\frac{1}{x} + \frac{x^3}{\sqrt{23}} \right) dx = -\frac{\alpha}{3(\alpha+1)} (3x^\beta + x^\gamma)^\alpha (\alpha+1) + C, x>0, (\alpha, \beta, \gamma \in \mathbb{Z})$, where C is the constant of integration, then $\alpha+\beta+\gamma$ is equal to _____.

Correct Answer: 19

Solution Analysis: $\int (\frac{1}{3}x^2 + \frac{1}{4}x^4)(\frac{3}{x} + \frac{1}{3}x^3)^{\frac{1}{23}} dx$ using $u = (\frac{3}{x} + \frac{1}{3}x^3) \Rightarrow$
 $du = -\frac{3}{x^2} + x^2 dx$
 $\int \frac{1}{3}x^2 + \frac{1}{4}x^4 dx \cdot \frac{1}{23} u^{-\frac{22}{23}} = \frac{1}{23} \int (\frac{1}{3}x^2 + \frac{1}{4}x^4) u^{-\frac{22}{23}} du$
 $= \frac{1}{23} \int (\frac{1}{3}x^2 + \frac{1}{4}x^4) \cdot \frac{1}{-22} u^{-\frac{22}{23} - 1} du$
 $= -\frac{1}{23 \cdot 22} \int (\frac{1}{3}x^2 + \frac{1}{4}x^4) u^{-\frac{45}{23}} du$
 $= -\frac{1}{23 \cdot 22} \cdot \frac{1}{-\frac{45}{23}} u^{-\frac{45}{23} + 1} + C$
 $= \frac{1}{23 \cdot 22 \cdot \frac{45}{23}} u^{-\frac{22}{23}} + C$
 $= \frac{1}{22 \cdot 45} u^{-\frac{22}{23}} + C$
 $= \frac{1}{990} (\frac{3}{x} + \frac{1}{3}x^3)^{-\frac{22}{23}} + C$
 $= \frac{1}{990} (\frac{3}{x} + \frac{1}{3}x^3)^{-\frac{22}{23}} + C$